

# 8 Rationalism: The Geometry of the Mind

The apology for rigid classification was first offered in Chapter 1 and has been repeated several times since. It is again in order, for “rationalism” is a term that has come to suggest so many meanings that nearly any objection to its application will stand on sure ground. It is customary, for example, to refer to the eighteenth-century “Enlightenment” as “the Age of Reason,” and its chief architects, needless to say, were all dedicated to reason. Yet, most of them—Voltaire, Hume, Diderot, D’Alembert, Condorcet—were empiricists, at least according to the broad definition given in the previous chapter. Even Rousseau, so passionately romantic and, in the common sense of the term, idealistic, was durably empiricist in addressing the major political questions of his time. And we have already observed that Locke’s empiricism was still able to embrace intuition and the possibility of moral maxims whose certainty was as great as those of mathematics. Nonetheless, there are fundamental differences between, on the one hand, Locke, Berkeley, Hume, James Mill, and J. S. Mill and, on the other, Descartes, Spinoza, Leibniz, and Kant. Moreover, a distinguishable assortment of “psychologies” has arisen from these differences. In this chapter, we will focus on two of the more important of the differences, the two that seem to be especially at odds and that played the greatest part by far in setting the infant science of psychology off on two separate paths. One difference is epistemological and center on the issue of innate ideas. The other is methodological and concerns the relative importance of rational versus empirical modes of inquiry and explanation. Naturally enough, the epistemological bias virtually determines the choice of method. Because these two issues are central, we might best begin by examining the meaning of each.

## Innate Ideas

Recall that Plato’s theory of knowledge was, at the root, nativistic. He argues in several dialogues that the eternal truths are locked within our souls, even

before birth, and that learning, properly conceived, is a species of recollection or reminiscence. In the *Theatetus*, Socrates makes short shrift of that marvelously Humean maxim, "Man is the measure of all things," and notes that, were the senses enough, dog-faced baboons could qualify as philosophers. The major portion of the Platonic legacy received by Aristotle, who was otherwise aloof to the theory of ideas, involved this skepticism toward the evidence of perception. Recognizing that the senses were responsive to changes in the material world, Aristotle too concluded that they were of little service in the search for the unchanging and universal principles. Accordingly, it was metaphysics that was to be the "first philosophy," and it was this ranking that Francis Bacon challenged.

But let us remain clear as to what stands behind the ranking. Aristotle was something of a "common-sense realist" in his psychological writings, not unlike Thomas Reid. He took for granted the essential accuracy and usefulness of the senses throughout the animal kingdom, for survival would be imperiled otherwise. Nature, as Aristotle said more than once, does nothing without a purpose. The senses, the perceptual and motor systems, the processes of memory and learning—all this is what serves the local as well as the long-range interests of the individual organism and the species as a whole. Thus, as fact-gathering instruments, the senses are essential and, except under peculiar conditions, faithful and accurate in their reports. However, what the evidence of experience can provide is precisely what calls for an explanation, not the explanation itself. We can observe, for example, that animal with hearts also have kidneys, but nothing in the observation will explain *why* this is so. Accordingly, scientific knowledge, which includes both facts and the explanation of these facts, must be grounded in nonempirical and irreducibly rational considerations. Ultimately, we have only understood something when we can supply the reason for its being what it is. This is where the "final causes" enter, and where the more empiricistic theorists politely (or not so politely) exit!

In the medieval period, as we have noted, contentions between adherents of each of these perspectives reappeared, this time in the form of the problem of universals. The eye can see one cat at a time, but the mind "knows" of the "universal cat." Such knowledge assumes a knowing faculty that is not sensory, and it is therefore a form of knowledge that cannot be given in experience. It is knowledge the mind must have prior to experience in order for experience to teach us anything. The words *given* and *prior* are at the heart of the problem of innate ideas. In Chapter 2, we reviewed Aristotle's objections to the strict (Platonic) nativistic account which seemed to require that infants enter the world knowing a number of things which, later in life, they will have so much trouble learning. Aristotle's jibe in this connection was the prototypic one and has been repeated in every generation by those seeking to disarm the nativist. The stock reply, of course, is that children do not have difficulty with the truth



but with the language they must learn in order to articulate it in the culturally approved form. And the rebuttal to this is that such so-called "knowledge" is only linguistic in the first place. This rebuttal, in turn, prompts the reply that were such truths merely verbal, there would have been no reason for every literate society in human history to have invented terms to express them. This is not a debate likely to end in victory.

Everyone tutored in arithmetic knows that there is no number so large that one cannot be added to it. No one knows this by experience, for no one has actually conducted the experiment and determined that the prediction has succeeded. To say that we know this by inference or generalization is to impute to the fact the less-than-certain status that attaches to all other inferences. (This once prompted Jean Piaget to quip that the radical empiricist believe the series of positive integers was discovered one at a time!) It is not merely likely that there is no number so large that one cannot be added to it. It is absolutely and irrefutably certain. If it is an inference, it is not the sort of inference generated by sensory knowledge or by experience. Since the fact is not established by experience in the first place and since it cannot be confirmed by experience, and, finally, since it is a fact, it must be one not given in experience. Thus, it is known a priori. That which could not possibly be given in experience is the backbone of claimed "innate ideas." Employed this way, the term "innate idea" does not require that infants be consciously aware of the fact or facts, only that, in their sufficient maturity, they will know certain things for certain. This knowledge will be the result of maturation alone, not instruction or experience. Advocates of this thesis claim only the existence of certain innate principles or archetypes of thought that assimilate experience and determine its psychological character. This is the sense of a priori that continues to enliven psychological research and theory. It is a concept of innateness not wed to any particular period of maturation and surely not infancy. We do not have to grant the infant the ability to add in order to know that it is not experience that proves that no number is so great that it cannot be greater. The rationalists to be studied in this chapter all subscribed to the doctrine of innate ideas so defined.

### Methodological Rationalism

Rationalists are further distinguished from empiricists by the method they recommend to discover nature's laws. As the term implies, rationalism is a commitment to thought, reflection, deductive rigor, an argumentative chain in which successive links are joined by the dictates of reason. The goal is a rational and intelligible account of why things are the way they are and not some other way. If the empiricist traditionally has been content to discover what is, the rationalist considers what must be. The empiricist's evidence has always been the data of experience; the rationalist's, the necessary proofs following

from axioms and propositions. Integral to the rationalist tradition has been mathematics and particularly geometry. Euclid's theorems have served more than one leading rationalist as the model for epistemology, even for reality. Recall the Pythagorean theory according to which the *tetrakys* (1, 2, 3, 4) generates respectively the point, the line, the plane, and the solid, such that reality itself is spawned by number. Logic, a kind of verbal geometry, has been the rationalist's method, standing in its own defense. If a skeptical attitude toward the senses has been added to this, it has been based on the conviction that there are abiding truths, immutable and deducible, which attributes or essences exist beyond the reach of the senses. A truth that cannot be seen must be present in the mind independently of experience. Thus, the rationalist does not reject the purely local and ephemeral facts of perception, but requires that they be incorporated within a logical system the truth of which is beyond sense but accessible to reason.

In its prescientific period, psychology was largely a debate between rationalists and empiricists. The great intellectual achievement, from Copernicus to Newton, was the marriage of these two perspectives. Science could not have been developed by way of a radical empiricism; nor could it have flourished if extreme forms of rationalism were adopted at the expense of systematic observation and classification. Aristotle had brilliantly combined the best features of empiricism and rationalism, but had subsumed both under a broader metaphysical system that conferred excessive authority on "final causes." The laws of science describe nature as it is experienced, not the way it "ought" to be. To the extent that Aristotle had preconceived notions about how nature "ought" to be, and to the extent that succeeding generations of rationalists were guided by his preconceptions, then, to these extents, the modern era begins with the rejection of the authority of Aristotle. Bacon, the empiricist, led the British movement away from Aristotle and his Scholastic following. On the Continent, the leading spokesman for the new era was René Descartes.

### René Descartes (1596–1650)

Descartes stands in the same relation to the Continental tradition of rationalism as Bacon does to British empiricism. Like Bacon, whom he mentions only rarely in his works and whose *Novum Organum* was probably unknown to Descartes as he began his own *Discourse*, the French philosopher saw unbridled skepticism as his most formidable adversary. In this sense, and notwithstanding the considerable differences between the Baconian and the Cartesian approaches to science, both philosophers share a common intellectual challenge: Renaissance skepticism and its tendency to breed superstitious alternatives.

We have already touched upon the skeptical strains of English scholarship in Bacon's time and we need not add much to the previous chapter in order to



describe the climate in France early in the seventeenth century. Copernicus's *De Revolutionibus Orbium* (1543) developed the case for the motion of the earth against Scholastic arguments to the contrary. Recall that Aristotle was quite willing to entertain the possibility, and that, in finally coming out against it, he accepted the geocentric alternative only weakly. Still, in the politically charged climate of the late Renaissance, distinctions between Aristotle and his Scholastic disciples were crudely blurred such that any successful attack on the latter was taken as a telling refutation of the philosopher as well.

In 1609, Johannes Kepler published the first two of his laws of planetary motion which required that planetary orbits be elliptical and, further, that the sun be the immobile center of the revolutions. In the same year Galileo had observed the moons of Jupiter with the aid of a telescope fashioned by his own hands. Astrology, theology, and much of the Hermetic corpus had an abiding commitment to a "sacred numerology" according to which "7" enjoyed a privileged place: for example, the seven days for creation, the seven days of the week, the seven sisters of the Pleiades, and the seven heavenly bodies. Galileo's observations demanded a recount. These developments, attending and following the Reformation, hardened the Church's stand against heresy. In Europe the consequences were greater and bloodier than in England. Bruno had been burned as a heretic in Descartes' lifetime (1600), and Galileo would be called before the Inquisitors in 1633, the year Descartes' *De Mundo* (which subscribed to the Copernican theory) could have been published had the author not held it back. Not long before Descartes' birth, wars between the Protestant Huguenots and French Catholics had ravaged the country. St. Bartholomew's Massacre (1572) had ushered in a year of butchery grim even by the macabre standards of religious persecution. Ramus, the sedulous anti-Aristotelian at the University of Paris, was one of the first casualties of the Massacre, and this tells us that, at least in France, the line separating philosophical and Roman Catholic orthodoxy was vanishingly thin.

Here, then, was the climate of Descartes' early development: growing achievements in science, the residue of Hermetic mysticism, the sullen skepticism and cheery materialism of the followers of Montaigne, the harsh, retaliatory measures of traditional authority, the numbing conformity of unthinking "Aristotelians" whom Aristotle himself would have abhorred. Descartes' rearing in this climate was traditional and upperclass, with his first instruction received at the hands of the Jesuits and emphasizing mathematics. From them he learned to respect learning itself, to devote himself to the ultimate purposes of Christianity, and to see in science, as it was known in his time, but another version of God's eternal wisdom and power. But from his own native genius, he came to doubt, as well. The great philosophical accomplishments of his life must be viewed as the triumphant reply to his own doubts. Even his lasting contributions to science and mathematics, his researches in optics and, especially, the optics of refraction, his application of algebra to geometry and con-

sequent founding of analytic geometry, even these are supplementary to and intended as proofs of his larger, philosophical system, his rationalism. Descartes' psychological theses are to be found in the *Discourse on Method*,<sup>1</sup> the second and the sixth of his *Meditations*,<sup>2</sup> and *The Passions of the Mind*.<sup>3</sup> His own summary of his broad and complex system was published in 1644, as the *Principles of Philosophy*.<sup>4</sup> His controversial and most influential psychobiological work, the *Treatise on Man and the Formation of the Fetus*, was published posthumously in 1664.<sup>5</sup>

Descartes' "method," which is summarized in Part II of his *Discourse*,<sup>6</sup> is really only a four-part maxim about the way the unprejudiced mind should go about its business: first, to accept nothing as true except that whose truth presents itself to the mind with such clarity and vividness as to remove the merest element of doubt; second, to divide a problem into as many discriminable elements as possible; third, to work from the solution of the smallest problem gradually up to the solution of the grandest; finally, to guarantee that the solution thus arrived at is sufficiently general as to allow no exception. Having achieved success in uniting geometry and algebra through this method, Descartes was persuaded that it might profitably be extended to all sciences. To aid him in this grander enterprise, he adopts several "moral" maxims: he will obey the laws of his country and adhere firmly to the Faith; he will adopt his provisional skepticism as if it were a proven certitude until the weight of reason requires that it be rejected; he will adopt a kind of stoic resignation toward those matters beyond the control of any individual while recognizing that his thoughts at least were within the perimeter of his powers.<sup>7</sup>

Applying the method epistemology, Descartes recognizes the limitations of the senses and adopts the skeptic's position that all is illusion and self-deception. Is there, then, anything that might justify our belief in it? Is it not just as plausible to assume that the entire fabric of apparent life is a fiction, an invention by nature assuring us of eternal ignorance?

Whilst I thus wished to think that all was false, it was absolutely necessary that I, who thus thought, should be somewhat; and as I observed that this truth, I THINK, HENCE I AM, was so certain and of such evidence, that no ground of doubt, however extravagant, could be alleged by the Skeptics capable of shaking it, I concluded that I might, without scruple, accept it as the first principle of the Philosophy of which I was in search.<sup>8</sup>

The skeptic whose doubts extend as far as his very existence finally is caught in a contradiction: that which doubts must be; that which thinks must be. Even if the body is an illusion, even if all our actions and experiences are unreal, the ideas of the mind must exist, or doubt itself is impossible. It is reason, therefore, that gives indubitability to existence, not matter. The perfected existence of triangle is rationally provided even if there is no triangle to be found and



representations of perfection. These must come from God, so God must exist.

even if a perfect one can never be formed materially. That the human mind can possess such perfected notions in an imperfect material world entails an immaterial author of perfection and this, of course, is God.

It is in the two concluding parts (V and VI) of the *Discourse* that Descartes reviews the many findings he has made in the biological and physical sciences using his method. He discusses the circulation of the blood and refers to the brilliant research "of a physician of England"<sup>9</sup> (William Harvey) for those who require experimental confirmation of that which reason can demonstrate. He also makes reference to unpublished treatises of his own which, because of their controversial elements, he has decided not to have published during his life. These include his essays in support of the Copernican system and, perhaps, his *Treatise on Man*. He also advises future workers to begin with reason and the immediate facts at hand before plunging forth into elaborate experiments, since the latter do not become necessary until knowledge is already well advanced. In this recommendation, Descartes conveys the essential flavor of the hypothetico-deductive method.<sup>10</sup>

Empiricism  
and exp.  
method  
Rationalism  
only reason  
exp. m.p.

Disputes between the ancient Greek atomists and idealists reflected an implicit disagreement over the status of mental as opposed to purely material realities. But Descartes was the first to put the mind-body problem in a form requiring the attention of scientists and philosophers alike. It may even be proper to say that the differences between modern materialism and ancient atomism—and these differences are very great—are the result of Descartes' analysis of the mind-body problem and the solution he offered to it. Psychological materialism, which in the nineteenth century became physiological psychology, is founded on the explicit rejection of both his method of analysis and his solution. Both, however, have proved to be quite durable.

It is useful to begin with Descartes' solution: mind and matter are of a qualitatively different sort, independent of each other; on no account is mind conceivably reducible to matter. Thus, Descartes is a dualist. He arrives at dualism, however, through the very skepticism to which his method commits him. He reviews his *cogito ergo sum* in the second Meditation and goes on to ask what sort of thing he is. He quickly dismisses Aristotle's definition of the human as a rational animal because this definition fails to tell us what is "rational" and what is "animal." He then reviews the various attributes he believes it has: body, hands, feet, hunger and thirst, and so forth, and a number of things it does; for example, walking, hearing, sleeping. All these, however, can be illusory. That is, the only necessary attribute, made necessary by the *cogito*, is thinking. Thus he must be a thinking thing.

I suppose there exists an extremely powerful and . . . malignant being . . . directed toward deceiving me. Can I affirm that I possess any one of all those attributes of which I have lately spoken as belonging to the nature of body? . . . The first mentioned were the pow-

ers of nutrition and walking; but if it be true that I have no body it is true likewise that I am capable neither of walking nor of being nourished. Perception is another attribute of the soul; but perception too is impossible without the body. . . . Thinking is another attribute of the soul; and here I discover what properly belongs to myself. This alone is inseparable from me. I am, therefore, precisely speaking, only a thinking thing, that is, mind . . . or reason.<sup>11</sup>

Matter is extended and is, therefore, in a place. Mind is unextended. That the soul's intellection comes to affect the body indicates that it must be in some sort of intimate contact with the body. This conclusion, introduced in the sixth Meditation, is the entire topic of the *Passions of the Mind*. Before turning to it, we should note an especially subtle point raised in the sixth Meditation representing that aspect of Cartesian philosophy that declares the existence of innate ideas. He accepts the empirical (Aristotelian) assertion that we possess a passive faculty of perception by which the external world impresses the senses. He does not doubt but that our knowledge of sensible things is created this way.

But this would be useless to me, if there did not exist in me, or in some other thing, another active faculty capable of forming and producing those ideas.<sup>12</sup>

The truths of speculative geometry are, we may concede, truths about actual figures but not truths that can impress the passive faculty of perception. Only reason can analyze the figures of geometry in such a way as to discern the general truths. He is even more explicitly Platonic in the *Principles of Philosophy* when he explains the failure of some to know these general truths in terms of their being taken in by the sensations of the body.<sup>13</sup>

Descartes' major psychological problem and the one that has vexed dualists ever since is that of accounting for the manner in which an immaterial, unextended agent (the soul) can influence an extended, material substance (the body). How can an idea move a muscle if the idea has no mass? The seeming impossibility of this has led some to adopt either a materialistic monism, insisting that all finally reduces to matter, or a mentalistic monism, insisting that all finally reduces to mind, or a neutral monism, taking no side in the issue except to insist that, ultimately, all must reduce to one or the other or some third, presently unconceived alternative. Descartes did not resolve the dilemma in *The Passions of the Mind*, but he subscribed to a position that veritably founded the dualist tradition. His analysis is as follows.

He reserves to the body all that can be imagined as pertaining to body: sensitivity, motion, extension, growth, decay. To the mind, however, he imputes that which is inconceivable in bodies: thought. He divides thought into that which impels voluntary action and that which is responsible for feeling.<sup>14</sup> Most



of our perceptions are the result of the action of external objects on the sensory nerves, and many of our feelings such as heat, cold, pain, and hunger are also referable to similar neural mechanisms.<sup>15</sup> Our emotional feelings, however, which can occur in the absence of any external stimulus, and our reflections on these, cannot exist in mere bodies and cannot, therefore, be attributed merely to our own bodies. These are in the province of the soul which, while not itself a body, is united as a principle to every part of the body.<sup>16</sup> When the body dies, the soul withdraws. But while the body lives, the will of the soul influences the body through its ability to regulate the flow of animal spirits from the brain to all the nerves associated with experience, action, and feeling. Descartes' best guess is that the site of this control is the pineal gland which, unlike the other structures of the brain, is not duplicated on each side and is located propitiously in the center of the brain.<sup>17</sup> Individuals differ in part because their brains differ,<sup>18</sup> but in all cases the passions result from the flow of the animal spirits contained in the cavities of the brain.<sup>19</sup> The soul's ability to direct the spirits is made possible merely by its willing to do so. Through God, this will is free.<sup>20</sup> He offers an illustration:

If we see some animal approach us, the light reflected from its body depicts two images of it, one in each of our eyes. The two images, by way of the optic nerves, form two others in the interior surface of the brain. . . . The images then radiate toward the small gland which the spirits encircle. . . . The two brain images form but one image on the gland which, acting immediately on the soul, causes it to see the shape of the animal. . . . The impressions which . . . a terrifying object makes on the gland causes fear in certain men, and yet in other men can excite courage and confidence . . . all brains are not constituted in the same manner.<sup>21</sup>

In this passage we have Descartes' introduction of the concept of the reflex which would engage the mind of eighteenth-century scientists throughout Europe. His own reluctance to include the soul in this purely materialist account of sensation and behavior would not be preserved by many of his outstanding successors in a more liberal period. Still, it is Descartes who first attempted to found biology on the same mathematical foundation that Kepler had given to astronomy. It was Descartes who insisted on placing the behavior of all animals and much of human behavior in the context of natural science. Moreover, it was Descartes who advised that we turn away from the search for Aristotle's final causes,<sup>22</sup> that we exercise reason instead of blind faith, that we place our trust in that which about us is beyond doubt: our own thoughts.

His direct contributions to psychology were impressive. That portion of his dualism that was materialistic is the cornerstone of modern neuropsychology. His influential writings on the reflex-connections between sensation and action,

began a line of inquiry that culminated in the research and theory of Ivan Pavlov. His *Method*, a version of which was shared by Galileo, saved science from that vain ritual of fact-gathering that Bacon's *Novum* seemed to demand. In presenting thought and certain kinds of feeling as the distinguishing features of human beings, he prepared the way for an experimental psychology of consciousness that would be inaugurated by Wundt in the nineteenth century.

Indirectly, Descartes' influence is felt in psychology through the effect his writings had on later critics in the empiricist tradition. It was Descartes' position on innate ideas that marks the point of departure between the (British) empirical and the (Continental) rationalist schools. But even Locke, in his discussion of intuition, imagination, the axiomatic nature of moral precepts, and the reality of ideas, even Locke the empiricist in his battle with the skeptics, borrows more from rationalism than he rejects. Descartes' place in the ranks of speculative science and psychology is at the front and durably so. He was not an antiempiricist; he was just not only an empiricist. He was not an antimaterialist; he was merely not only a materialist. This is why it has been far easier to challenge one or another of his assertions than it has been to escape his presence even now, three centuries since his death.

As we shall see in the next chapter, Descartes' total effect on thought took the form of an elaborate "Cartesianism" against which the eighteenth-century *philosophes* fought so tirelessly. At a superficial level, the war on Cartesianism usually centered on its physics which, in the wake of Newton's achievement, was judged to be wrong, if not ridiculous. In comparing Newton and Descartes, the *philosophes* argued that the chief difference was one of *method*. Newton, they thought, succeeded because he had relied on the evidence of direct experience. Descartes failed because he placed his trust in the tidy rationalism of "Schoolmen." Moreover, Newton himself had said, "Hypotheses no fingo" ("I frame no hypotheses"), a phrase that made it difficult for the eighteenth-century commentator to recognize the fundamental difference between Newton and Bacon in the matter of philosophy of science. Locke, Reid, Hume, and most of the French empiricists explicitly paired Newton and Bacon when discussing the proper method of philosophizing. On this account, science and philosophy are the same: They seek the truth by confining themselves to observation and experiment, and by resisting the temptation to *rationalize* nature.

As we know, however, Newton framed a good many hypotheses, and Descartes made a good many observations, including experimental ones. At the heart of eighteenth century anti-Cartesianism was not so much a rejection of Descartes' methods as an opposition to a number of his conclusions. What, after all, follows from claims of the sort, "all brains are not constituted in the same manner," if not a direct challenge to claims of the sort, "All men are created equal"? Does not the same reasoning lead to notions of the king's "di-



vine right” and to justifications of classes and castes? If, as Descartes’ psychology holds, the human soul is indestructible and immaterial, and if the distinguishing feature of the human mind is based upon something that transcends experience, on what basis can the historic authority of Church and Crown be challenged? Note that “Cartesianism,” to the eighteenth-century reformer, appeared to be something vastly more formidable than a theory of physics or a method of speculation or a system of metaphysics. It was explicitly a theory of human psychology and, as such, a theory of government and society. Although played out as a battle between “Newtonians” and “Cartesians,” the intellectual collisions of the eighteenth century were between Whigs and Tories.

### Descartes and the Animals

Those who have expressed a principled concern for the manner in which scientists have used and, yes, abused animals tend to place much of the blame on Descartes’ psychological writings in which nonhuman animals are described as “automatons,” or a kind of machine. It is true that such a position can be found in Descartes’ writings, though it is also clear that he accorded nonhuman animals the full range of perceptual and (even) motivational and emotional processes. Assuming that they lacked all rationality, he was led to the conclusion that they were unable rationally to interpret stimulation and therefore were not conscious in the relevant respects. It goes without saying that here, too, the authority of reason—when it trumps the direct evidence of daily life—is worth resisting at every turn.

### Benedict de Spinoza (1632–1677)

Spain’s wars with English and the defeat of the Spanish Armada in 1588 not only loosened the hold of the Roman Church throughout Europe but also made it difficult for Spain to continue to impose orthodoxy at home. Many of the Jews who had been forced by the Spanish Inquisition to adopt Catholicism now either began to rebel openly or to find their way out of Spain and into states eager to accept non-Catholic citizens. Holland was especially tolerant and it was to there that the Spinoza family migrated toward the end of the sixteenth century. Spinoza was thus reared as a Jew in the Christian city of Amsterdam at a time when the “new philosophy” (i.e., Descartes’) was all the rage in intellectual quarters. What he borrowed from this new philosophy was sufficiently unorthodox to result in his excommunication from the Jewish community. What he assimilated from his training at home and in the synagogue was enough to exclude him from the circle of important Christians. And, in finally granting to intuition a status even higher than that accorded to reason, he was unable to maintain a position among the intellectuals of the Enlightenment. He

was something of a Maimonides among scholastics, a “naturalist” among mystics, a situationist among absolutists, and an absolutist among skeptics. Spinoza lived a **very difficult life**.

Perhaps the best approach to Spinoza’s psychology is by way of one of Blaise Pascal’s *Pensées*:

I cannot forgive Descartes. In all his philosophy he would have been quite willing to dispense with God. **But he could not help granting Him a flick of the forefinger to start the world in motion; beyond this, he has no further need of God.**<sup>23</sup>

Descartes’ system of philosophy took for granted the separate and independent categories of God, matter and mind. While it had not suggested that the last two could exist without the first, it was **implicit in Descartes’ rational psychology that, once God’s forefinger flicked, the balance of nature and natural events could be studied rationally**. We have already described his **impatience with the search for final causes** and his **resulting concerns with efficient and material causes**. These attitudes translate into one kind of Cartesianism, the kind that grants divine authorship in accounting for the world but restricts scientific and philosophical inquiry to methods and events with, we might say, theology held constant. **The cosmology here reserves to God the creative act of bringing everything into being—and then letting it run its course according to the immutable laws of science**. If divine intervention ever takes place again, it must result in the suspension of these very laws, the result being nothing less than a **miracle**. The *Deists* of the eighteenth century would be more transparent in adopting this perspective, but it was highly risky in the seventeenth.

Unlike Pascal, Spinoza was not influential in his own time although Leibniz, despite his politically motivated protestations to the contrary, seems to have been impressed by a number of arguments appearing in Spinoza’s *Ethics*. It would be in the nineteenth century that interest in Spinoza would sharpen, chiefly among the German Romantics who often found much to condemn in Spinoza’s deterministic philosophy. Lessing, Schelling, and Fichte all found occasions for discussing Spinoza. But we pause to review **Spinoza’s psychology** briefly here because it **exposes the sensed tension between reason and passion, between free will and determinism, which has been so much a part of modern psychological thought**. It also reflects an awareness that science will either incorporate God completely into its deliberations or must exclude him **just as completely**. Spinoza is important, then, as one who saw clearly how new the new philosophy was.

Like Descartes, Leibniz, and Pascal, **Spinoza was well versed and active in science**. He worked as a lens-maker and had the fullest grasp of the principles of optics, which were, of course, geometric. Like Descartes and Leibniz, he was **convinced that there was certain truth in a world of apparent change**. Un-



like Descartes, he could find no reason in logic or experience to assume that mind, matter, and God were to be relegated to distinct categories. It followed that if God is the author of all, His presence must be in all, since "things which have nothing in common cannot be one the cause of the other."<sup>24</sup> Thus, if God is the cause of all things, it makes no sense at all to talk about human freedom. Our very knowledge of good and evil is sufficient to prove that we were not born free, for, if we were, this constraining knowledge would not exist.<sup>25</sup> Our "freedom," therefore, is of a different sort. Since God is a "thinking thing," our thoughts will either share in His or be imperfect. If imperfect, our actions will be compelled by passion rather than emotion. The distinction between the two is important. For Spinoza, a passion is a feeling toward that about which we have no clear idea, whereas an emotion is a feeling shaped by a distinct idea. So-called "blind rage" is an instance of passion, while love for our fellows is an emotion.

This is a quite interesting distinction, the kernel of which can be found in parts of Aristotle's ethical writings and more fully in Hellenistic Stoicism. To respond passionately is to respond intemperately, which is to say in a manner not constrained by that knowledge and those principles that impose proportion and balance on our conduct. One of the grounds on which Aristotle withheld ascriptions of virtue from nonhuman animals was their presumed lack of a deliberative power. As their choices were assumed to be based not on deliberation but on a kind of passion, there was no controlling principle involved and, therefore, nothing of virtue in the act. To act in a virtuous way, therefore, calls for deliberation; it also calls for commitment and resolve. Thus, in Aristotle too there is room for an affective component to which we must be properly disposed. For example, we must be disposed to be angry over injustice, not justice.

Stoic philosophers, especially in the early Christian period, were more radical in their theories of the emotions. They took them to be something of a disease or malady. The actor impelled not by reason but by passion is clearly suffering from a disordered mind; *literally* disordered in that the right order of things—the rule of reason—has been upset. Spinoza's thesis is indebted to these older traditions in philosophy. If we have a clear idea of whatever it is before us, then our actions toward or in response to it will not be the result of compulsion but of commitment. Our actions will not be impelled by passion, but given resolve by emotions that are under the mind's control.

But what does it mean to have a clear idea? The clear ideas is none other than a rational awareness of the fact that whatever is the case is what it is necessarily. "The mind," says Spinoza, "has greater power over the emotions and is less subject thereto, in so far as it understands all things as necessary."<sup>26</sup> Thus, where Descartes' conception of body and soul (mind) required each to have some vague kind of relation to the other, Spinoza radically divides the psychological and the spiritual: learning, perception, memory, and emotion re-

quires a body and end with the end of the body.<sup>27</sup> There is an afterlife only in the sense that God, as a thinking thing, eternally retains the idea of the essence of the given individual.<sup>28</sup> Since this essence is mind itself, mind is eternal.<sup>29</sup> As he argues in his *Proposition IV*, nothing can be destroyed except by something external to itself. To the extent that the mind is possessed of its own adequate ideas and is not totally passive (i.e., acted upon from the outside), mind as such is indestructible.

To this point Spinoza's thesis is not unlike St. Augustine's and even has Berkelean features. But when he turns to matters of direct, psychological consequence, his philosophy becomes radical. Rationalism becomes a kind of emotionalism. What we call "good" and "evil" he says are "nothing else but the emotions of pleasure and pain."<sup>30</sup> But pleasure and pain are, respectively, the mind's awareness of its strengths and its weaknesses.<sup>31</sup> It is the mind that seeks to endure eternally and it is the mind that knows that this is possible only through its own activity, which is to say through its clear ideas. Passion, as a passive state, perception as a fleeting affair, and mere imagination as the repository of purely contingent events will not permit the mind to endure. The mind equipped with these is aware of its weaknesses, fearful of its transitoriness, and thus pained. Reason, however, and, more important, intuition, bring to the mind the clear idea of necessity, whose possession is pleasure. The emotion so created is the most intense, since "an emotion toward that which we conceive as necessary is, when other conditions are equal, more intense than an emotion towards that which is possible, or contingent, or non-necessary."<sup>32</sup> And, as an emotion can only be affected by another emotion (*Part IV, Prop. VII*), we begin to see that the rationally achieved adequate idea is the only basis upon which emotions might come to be attached to the right objects of thought.

Spinoza's deterministic psychology is unreserved:

The mind is determined to wish this or that by a cause, which has also been determined by another cause . . . and so on to infinity.<sup>33</sup>

The ultimate or first cause is God. To the extent that the mind is fixed on "adequate ideas," which are the ideas of necessity, it is active; otherwise, it is passive. When active, it endeavors to persist in its being, which is to say it strives for pleasure. Since this pleasure is the possession of adequate ideas and since the body is no such necessary entity and, finally, since things that have nothing in common cannot be the cause of one another, "body cannot determine mind to think."<sup>34</sup> However, it is in our nature to include the idea of our bodily existence in all of our ideas and, therefore, the mind is threatened by that which threatens the survival of the body. Intuitively we labor to preserve the activities of our bodies because of this sensed connection between bodily and mental survival. Since we believe that a threat to the body entails a threat to the activity of the mind, we strive to we repress those ideas of the body's injury or demise.<sup>35</sup>



The categories of Spinoza's psychology are passion, emotion, reason, and intuition. All are determined and, to that extent, the will is not to be described as free. What makes us unique, when we are unique, is the presence of clear or adequate ideas about the necessary causes of things, and these ideas lead back inevitably and inexorably to God. All else involves passive states of mind which, in the limiting case, would be eternal death. We are so constituted that our lives are devoted to perpetuating the existence of mental activity. The motive, at a superficial level, appears in the form of an egoistic quest for pleasure and aversion to pain. The unenlightened are moved in these respects by opinion rather than reason, and as a result of opinion and the laws of association they come to identify bodily death as the ultimate pain. Benignly, God has planted in us a higher sense, an intuitive awareness that is the active and impassable mind. But even when this mind grasps the essential nature of what is good, the emotions cannot be restrained by the truth of the knowledge, only by another and contrary emotion produced by such knowledge.<sup>36</sup> In this aspect of his theory, Spinoza stands in agreement with the British sentimentalists, with Hume, and with Kant: Reason alone will not produce moral conduct except to the extent that the rational exercise leads to or is correlated with the proper feeling. Our will is not the free cause of our actions but the necessary cause.<sup>37</sup> Given the will, action follows necessarily. Given God, the adequate idea, which is the idea that refers to Him, follows necessarily. This is worth rehearsing. What Spinoza requires is a means by which activity is initiated. Reasons for acting are not direct causes of action, for one may have a reason for doing something and still not do it. But a reason for acting may engender a certain desire of the will, and it is this that impels the action. Thus, given the will, there will be activity. But the will is not free, for it is engaged necessarily by prior events. Nor is the resulting action free, for it is impelled by the will. If the action is to be of moral worth, it must be traced back to an adequate idea, this being the necessary consequence of God.

In his uncompleted essay *On the Improvement of the Understanding*, Spinoza embellished his theory and offered a number of remarkably modern notions of learning and memory; for example, that memory is enhanced or degraded by the contextual features of the material to be memorized, that memory suffers when materials similar to those memorized are subsequently committed to memory, that memory is a brain process, and that every idea must have a correlate in the real world.<sup>38</sup> It is here also that he makes the traditional rationalist distinction between mere sensations or perceptions and the active assimilation of experience by the intellect. Of the various modes of knowing, only opinion is inferior to mere experience. The highest mode is that by which the essence of a thing is known. We know the essence of a circle, for example, when we know that it is the line resulting when a stylus is fixed at one end. To know the essence of a thing is to know what must follow from the fact of its

existence. To know an eternal truth is to know that, if it is true at all, its contrary could not be true. Thus, the Pythagorean theorem is an eternal truth in that there could not be a right-angle triangle the square of whose hypotenuse does not equal the sum of the squared sides. To know such truths and, indeed, to know even simple facts, is simultaneously to affirm and to negate. That is, every determination entails a negation. Once we have determined that Smith is an old man, we have, in this act of determination, denied that he is young, that he is female, that he is aluminum. To affirm is to limit. To define is to confine. The infinite, then, is either undefined (which Spinoza rejects) or is self-defined. To know an eternal truth is, therefore, to know the sense in which it is self-defining, self-causing. It is to know God.

Spinoza's "God," however, is neither the Yaweh of the Old Testament nor the humanized first cause of Christian Aristotelians. Rather, it is the god of the logicians; something of a deduction stripped of theology. His predictable standing as an atheist still receives scholarly attention, though there seems little doubt but that he regarded the necessitated transactions of the cosmos to be necessitated by more than a formal argument. Spinoza did not speak to his age even if, in a peculiar way, he spoke for it. His recourse to associationistic principles of learning and memory was hardly original, and his egoistic theory of motivation, while infuriating, had been in the wind since Hobbes, and since Montaigne before him. In striving for a unique form of monism, he offered something of a pantheism. God, as a thinking thing, inspires certain substances in a way that makes them thinking things too. Such a pantheism had to be judged either as heresy or lunacy, and Spinoza's contemporaries were willing to offer both judgments. Not until the nineteenth century would his philosophy be taken seriously by a major philosopher and, in taking it seriously, Hegel would reject a good part of it and force the rest to conform to his own system.

To Spinoza, also, we owe the insistence that philosophy and theology are different enterprises, that the former must be concerned with truth, no matter where it leads, while the latter must require reverence and a degree of compliance. Further, philosophy is a search for truths pertaining to nature, and God is known in nature through the laws which are (therefore) necessary. We, as thinking things driven to create pleasure in the mind, are constrained by nature to strive toward our essence and to be pained by its denial. The goal of human psychology, then, is self-actualization. In this, Spinoza serves as a model for a number of twentieth-century humanistic psychologies.

Though not a disciple of Descartes, (openly disagreeing with him, for example, on the issues of free will, dualism, and the division of nature into compartments), Spinoza shared with Descartes a realization that Greek wisdom was not enough, that Christianity had to be more than Aristotelianism, that a rational creature, stripped of all superstition and unbiased in the face of nature, could know the truth. His rationalism was a naturalism without becoming a



materialism. Spinoza did not doubt matter as did Berkeley, nor did he doubt mind as radical materialists did. He began with God and from that starting point he examined thinking matter. Implicitly, he defended materialism as applied to matter, since only materialism applies to that which is only matter; just as only idealism applies to a thinking thing when it is the thinking we wish to understand. What Spinoza was able to avoid through his complex metaphysics was the sort of trouble awaiting the usual varieties of dualism and materialism. Before Spinoza, philosophers by and large spoke in behalf of mind or matter, or mind and matter. Spinoza, however, was able to deduce from his major axioms the conclusion that mind and matter are the same substance viewed from different perspectives, thus introducing a quite subtle form of "double-aspect" or even "multiple-aspect" theory as a solution to the mind-body problem. At the most general perspectival level—say realism—we are prepared to claim that every existing thing triggers in us an idea of it by way of sensations, images, memories, and the like. From another perspective—call it idealism, since we are conscious only of sensations and ideas—we are obliged to accept these sensations and so forth as the only constituents of reality. From still another perspective—call it materialism, since knowledge can only be about things, and since these can enter the realm of thought only through their material effects on our material organs—reality consist only of matter.

Spinoza treats these claims as the product of certain habits of talking about truth, not as truths themselves. In their place he would offer a monistic theory according to which ideas and objects are substances of the same kind. For every object, there is a matched idea; for every idea a matched object. To have an idea, after all, is to have an idea of something. But to have an idea of what finally are only the attributes of a thing (an idea that rises no higher than the level of mere perception) is to have an incomplete and unclear idea. To know the object requires that we know the substance. The latter is what remains of a thing when its attributes are peeled away—that which allows it to be what it is eternally and immutably. Note that the idea of this is necessarily an imperishable idea, since for every object there is a corresponding idea. And it is in just this sense that the mind seeking to endure must be the mind seeking the adequate and clear idea. *—but one get ideas, they are not clear, so we will endure*

Spinoza's eternal substances are of one kind, though they display a mental and a physical "aspect." They do this not because of any fundamental duality in the universe, but because we have not thought of them clearly. And by the same token, we speak of God and Nature as if there were two distinct entities involved, failing to recognize the implicit contradiction. To ascribe to God all the properties of perfection, omnipotence, and omniscience, and then to argue that such being, at a determinate time, brought the world into being, is nonsense. For God to have created the universe, it would have been necessary for God to lack something and to attempt to remedy this through an act of creation.

But if this were true, God would not have been the perfected and universally inclusive being of the Hebrews or the Christians. To make theology coherent, then, Spinoza thought it was **logically necessary to take all substances—all substantial thing/thought items—as eternal in their substantial unity**. On this construal **there can be no essential difference between God and nature, mind and matter, or science and philosophy**. Goethe and Coleridge, as leaders of the nineteenth-century Romantic movement, both acknowledged their debts to Spinoza, and both would give impetus to that “Religion of Nature” which is the essence of Romanticism. Others would discover in Spinoza’s essays a new justification for the old-time religion; still others, for the new-fangled agnosticism. **As with Plato and Aristotle, Spinoza has something for everybody, but not always what the recipient thinks is in the package!**

### Gottfried Wilhelm von Leibniz (1646–1716)

Leibniz was only four years old when Descartes died. He lived in an intellectual climate dominated by Descartes and the “Cartesians,” a climate rapidly becoming as uniform in its reverence for Descartes as the world of letters once had been toward Aristotle. Leibniz’s father was professor of philosophy at Leipzig, and, while both the father and mother were dead before he completed his university studies, Leibniz may be said to have been blessed with a stimulating early environment. **He was exposed not only to the Greek and Latin classics but to the modern works of Bacon, Descartes, and Galileo**. Visits to London (brief) and Paris (four years) further exposed him to the most important ideas of the time. He met with Malebranche, the great Cartesian; he studied Pascal’s treatises in mathematics and invented a calculating machine even better than Pascal’s; through Huygens, his interest in optics was aroused; and Hobbes’ works enhanced his abiding concern with law, as the recently ended Thirty Years’ War engendered a passion for peace and tolerance. By way of introduction, we should also point out his discovery of differential calculus independently of Newton, and the bitter claims of priority that sounded across the English Channel for the better part of a decade.

Louis XIV, the Sun King, had become king of France three years before Leibniz was born and died a year earlier than Leibniz. Thus, **in addition to his own genius, his good head start, and the stimulating intellectual giants of his time, Leibniz passed his years in what were among the most self-consciously cultivated and achievement-oriented decades in modern history**. We might underscore this by listing some of the authors whose works were published during the reign of Louis XIV or who were still alive at the time: Hobbes, Locke, Descartes, Newton, Pascal, Spinoza, Gassendi, Leibniz, Malebranche, Huygens, and Moliere. Galileo had been dead only one year before Louis XIV acquired the throne. When the king died Voltaire was already nineteen and



Hume's *Treatise* was only twenty years off. Of those on this list who were Leibniz's contemporaries or who just preceded him, Leibniz agreed with none. His role in the history of ideas is secure, his influence diffuse and recurrent. His specific contributions to psychology, while not as great, are revealing and are presented most clearly in his disagreements with Locke and with Descartes. In rejecting Locke's empirical psychology, he reasserted the cognitive, highly mentalistic, and genetic character of human knowledge and feeling. And, in opposing Descartes' dualism, he rephrased the mind-body problem in a way that would render it more interesting to the modern era. We will begin with his anti-empirical arguments.

Locke's *Essay on the Human Understanding*, published in 1690, came to Leibniz's attention in 1688. He began to draft a rebuttal straight-away, although this rebuttal, *New Essays on the Understanding*,<sup>39</sup> did not appear until 1765. The reason is that Locke died the year Leibniz completed the work (1704), and Leibniz was not willing to argue with his ghost. Accordingly, the *New Essays* became just one of many Leibnizian contributions not made generally available until after the author's death. Like the rest of his work, the *New Essays* continues to influence the reflective mind.

Locke had begun Book II of his *Essay* with the central claim of all empiricists before and since:

Suppose the mind to be, as we say, white paper, void of all characters, without any ideas; how comes it to be furnished? Whence comes it by that vast store, which the busy and boundless fancy of man has painted on it with an almost endless variety? Whence has it all the materials of reason and knowledge? To this I answer, in one word, from experience.<sup>40</sup>

Leibniz's reply, a reply offered by all rationalists before and since, is that only some *thing* can be said to have an experience, and such a thing must be a mind somehow prepared to have experiences of a given sort. He identifies Locke's position (incorrectly) with Aristotle's, and attributes to Aristotle a statement also attributed to him by Duns Scotus, although we search in vain for this statement in any of Aristotle's works: "*Nothing is in the intellect which was not first in the senses.*" To this Leibniz replies, "*Nothing except the intellect itself.*"<sup>41</sup> He follows the introduction to his *New Essays* with a dialogue between Philalethes ("friend of sleep") and Theophilus ("friend of God"),<sup>42</sup> who are, respectively, the empiricist and the rationalist. After giving Philalethes the lines from Locke's *Essay* (quoted above), he has Theophilus offer the Leibnizian position on the problem of knowledge and the solution:

This *tabula rasa*, of which so much is said, is in my opinion only a fiction which nature does not admit. . . . Uniform things and those

which contain no variety are never anything but abstractions, like time, space, and other entities of pure mathematics. There is no body whatever whose parts are at rest, and there is no substance whatever that has nothing by which to distinguish it from every other . . . those who speak so frequently of this *tabula rasa*, after having taken away the ideas, cannot say what remain. . . . Experience is necessary, I admit, in order that the soul be determined to such or such thoughts, and in order that it take notice of the ideas which are in us; but by what means can experience and the senses give ideas? Has the soul windows, does it resemble tablets, is it like wax?<sup>43</sup>

Experience is necessary, on Leibniz's account, in order that the soul take notice of the ideas that are in us. What experience provides is a context for our thoughts, a direction for our ideas, a means of aiming our attention, disposing ourselves to action of a certain kind. It is impossible that an experience will produce an idea for the simple reason that an experience involves the physical confrontation of matter and the organs of sense, and an idea has nothing to do with these mechanical transactions. Perception, however, which is not a mere experience and which assumes a rational, attentive mind, can lead to ideas; but perception is not mechanical and cannot be reduced to the mechanical.<sup>44</sup> It is precisely on this basis that Descartes' dualism is to be rejected. Leibniz is clearest on this in his *Monadology*, written two years before he died and presented as a condensation of his metaphysics:

Moreover, it must be confessed that *perception* and that which depends upon it are *inexplicable on mechanical grounds*. . . . And supposing there were a machine, so constructed as to think, feel, and have perception; it might be conceived as increased in size, while keeping the same proportions, so that one might go into it as into a mill. That being so, we should, on examining its interior, find only parts which work one upon another, and never anything by which to explain a perception.<sup>45</sup>

Perception is a uniquely *psychological event*. It is that of which we are conscious. It is *qualitative* in a way that no purely quantitative (i.e., material) phenomenon can imitate. As we walk through the great mill of the mind, observing the spinning wheels and crashing hammers, we find nothing by which the mill could have a perception; not that the mill does *not* have such perceptions or is not aware of itself but that *nothing in its moving parts could convey as much*. We will return to Leibniz's mill in the last chapter.

The mind-body interactionism advanced by Descartes thus seemed at once confused and meaningless to Leibniz. Mind, on the Leibnizian account, is a simple substance, a *monad*, not reducible to anything, not deriving its character from any source beyond itself, not extended. As with all simple substances, it



is to be understood as a quality and not a quantity. In Leibniz's terms, its essence is *intensive* rather than extensive. The example of a point in mathematics is illustrative. A point is not a very small line or a very small fraction of a line. It is the idealized limit as extension approaches zero. Similarly, as quantity is stripped of its extensive features, there is a limit beyond which further reduction is not possible. This limit constitutes a quality of being, not a magnitude or extension. The limit of the body is also a simple substance; that is, a *monad*. Body as it is perceived is a composite whose extension arises from the assembly of simple substances. No two simple substances are alike. Each monad not only has a distinguishing quality but is the very "unit" of quality. Being dimensionless, it is not subject to modification from without. It has no window through which an external agency might enter and change it. In this respect, it makes no sense to speculate on the kind of interaction taking place between "mind" and "body" since each, properly conceived, is unique, independent, and ultimately unextended. Body and mind coexist and the relation between them is not causal but harmonic. If a note is loudly sounded in the presence of two resonators, we do not ask which of the resonators has established the sympathetic vibrations in the other. The two resonate in parallel as a result of being so constituted that, in the presence of the proper stimulus, each satisfies the conditions of its nature. So too with mind and body. Each, according to a pre-established harmony, exists compatibly with the other. The action of one is not caused by the other, as the mechanistic account requires; nor is the action of each reconciled to that of the other by some external "timekeeper" who occasionally steps in to make sure that all the clocks are on time—the view advanced by the philosophic occasionalists.

The universe is a collection of simple substances for which harmonic associations were established prior to their coming into being. Harmony is God's modality. If the monad is to change, it can do so only through an inner principle.<sup>46</sup> It is this internal principle, involving as it does a system of relations within the monad, that constitutes perception. This, however, differs from apperception or consciousness by which we not only perceive but are aware that we perceive. To the extent that every monad has an internal organization, it is perceptive. The monad whose internal principle allows both memory and perception may be called a soul.<sup>47</sup> It is clear from this that animals have soul. However, they do not have rational souls (i.e., minds), because, while they are able to perceive and even to retain the trace of former, consecutive perceptions, they are unaware of necessary truths. Human beings, too, insofar as their perceptions are simply united by memory, "act like the lower animals, resembling the empirical physicians whose methods are those of mere practice without theory. Indeed, in three-fourths of our actions we are nothing but empirics."<sup>48</sup> It is only in our knowledge of a rule, of a necessary relationship, that we display the uniquely human quality of human life.

More than any previous philosopher, Leibniz was concerned with and wrote

Unextended - occupies no physical space  
(unextended in time)

about an issue that would be at the core of modern psychology's problems, the unconscious. This is not to say that, in any complete sense, he anticipated Freud. Leibniz's use of the concept had little to do with the causes of motivation and nothing to do with psychopathology. He employs the concept, instead, in support of his position on the indestructibility of monads, on the distinction between perception and consciousness, and on the difference between the bare monad and the rational mind. Even in a dreamless sleep, he maintains, the monad does not perish (for it can't) and, therefore, since it also cannot exist without being affected in some way, perception, by definition, takes place. However, we are not aware of this perception because it is not accompanied by memory.<sup>49</sup> A number of unconscious (insensible) perceptions, stored in the mind, can add up in such a way as to break into consciousness. Indeed, there is a gradual scale separating the sleep of death from heightened awareness. We pass from one to the other in small steps, one of which constitutes a threshold. Moreover, we retain all that has happened to us even though we might not be actively aware of much of it. The representations of the past remain in the mind, creating "an influence greater than people think. . . . The present is big with the future and laden with the past."<sup>50</sup>

As with Spinoza, it is not easy to carve out a niche for Leibniz in the history of psychology. As the dual enemy of empiricism and materialism, he cannot be located in that philosophical tradition that led to the installation of psychology as an experimental science. His writings make it clear that he believed one could deduce most of what our contemporary experimentalists are searching for, and what is not easily deduced is either trivial or readily available through common experience. We have already acknowledged his attention to the unconscious and his formal introduction of the concept of subliminal perception. That division of experimental psychology devoted to sensory thresholds has a debt to Leibniz here, but the debt is several times removed. Even his challenges to Locke were hardly more than what one can find in the *Protagoras* or the *Meno*, and Leibniz is the first to recognize that he has taken the Platonist position in the dispute.<sup>51</sup> His discussions of the unity of consciousness, of the role of memory in consciousness, and of the differences between consciousness on the one hand and both perception and memory on the other, would surface again and again in both the theoretical and the experimental psychology of the late nineteenth century. Perhaps his most substantial direct effect on psychology—and, we might suspect, an unforeseen one—resulted from his telling criticism of Cartesian dualism. In illustrating its defects and contradictions, he did much to overturn Descartes' authority and to liberate thinking to the point of allowing an uncomplicated physiological psychology. Leibniz would not have applauded the reduction of mentalism to materialism—he specifically opposed it—but others would dismiss his caveats and focus instead on his successful refutations of Cartesianism.

Unlike Spinoza, Leibniz did little to restore idealism to a position of philo-



sophic significance, and, therefore, we are not even able to relate him to that tradition leading up to Hegelianism. But his emphasis upon activity and unity, the two abiding features of all and every simple substance (including mind), would reappear in the psychologies of Brentano, James, the Gestalt school, and even early behaviorism. His monism would, as we have noted, inspire confidence in those who might examine brain in order to unearth mind. His attribution of soul to animals and his insistence on the continuous evolution of various levels of organization and relation would surely not retard the development of an experimental psychology of animal intelligence. His unswerving commitment to the presence of innate characteristics and his logical arguments in the behalf of the necessity of *a priori* dispositions of the mind would constitute a starting point for one of the most influential philosophers of all time, Immanuel Kant.

### Immanuel Kant (1724–1804)

The position in which philosophy, metaphysics, and science were left by Hume's *Treatise* was anything but reassuring. Rationalism was rendered devalued in its singular goal, the search for eternal truths. Epistemology was reduced to psychology, and an associationistic psychology, at that. What the *Treatise* denied was that necessity could be proved to exist in nature. It denied that logic confirms such necessity and that the senses ever perceive it. It affirmed the existence of subjective necessity as a habit of the mind. We judge B to be the effect of A when the two occur together in space and time, when A always precedes B and these perceived unions are constant. Since experience alone is responsible for our belief in causation, and since, in principle, it is "possible for all objects to become causes or effects to each other," we are forced to acknowledge that "anything may produce anything."<sup>52</sup>

Hume did not deny that events were caused. Rather, he insisted that our commitment to this view cannot rest upon any basis other than experience and, as such, the view can never acquire the added baggage of necessity. It may happen that A is always followed by B, that no one has ever recorded an exception, that the interval between the two is perfectly constant. Still, all we know is A and B. "We are never sensible of any connection betwixt (them)."<sup>53</sup> We have knowledge of the events and knowledge of the temporal connection between them. We have no knowledge of necessity. That is, experience will confirm only A . . . B; not A . . . necessarily . . . B. On the same basis, moral distinctions are not derived from reason but from experience (and the feelings resulting from it).<sup>54</sup> And, again on the same basis, there can be no logically compelling argument against those who assert that the rational faculties themselves are but the outcome of natural material forces. For even though thought and matter seem to be different, experience suggests that "they are constantly united; which being all the circumstances, that enter into the idea of cause and

effect, when apply'd to the operations of matter, we may certainly conclude, that motion may be, and actually is, the cause of thought and perception." <sup>55</sup>

In a word, the *Treatise* removed moral precepts from the domain of the rationally deducible, removed necessity from the domain of cause and effect, and removed reason itself from the domain in which we locate the determinants of knowledge, feeling, and conduct. A rational philosophy designed to unearth necessary moral prescriptions must fail. A rational philosophy seeking to penetrate what *must* occur in nature must fail as well. The *Treatise* stripped natural science of "must" and moral science of "ought." Empirical psychology was all that survived.

If we are to appreciate the lasting contribution Kant made to psychology, we begin by examining a question that has tired philosophers for nearly two centuries: What was Kant's answer to Hume?<sup>56</sup> The answer is to be found in his towering achievement, *Critique of Pure Reason*,<sup>57</sup> in his summary and clarification of this work, *Prolegomena to Any Future Metaphysics*,<sup>58</sup> and in his *Groundwork of the Metaphysics of Morals*.<sup>59</sup> An implicit theory of psychology runs through every chapter of each of these, and often the theory is explicit as well. In the following discussion, there is little chance that all of Kantian philosophy will be penetrated, but the psychology in this philosophy will be evident.

All the major philosophers of the eighteenth century were quick to note the distinction, a distinction made by Plato and by a good number of scholastics, between propositions that seek to add to our knowledge about a thing and those that only assert a semantic identity. When we say, for example, that a body is an "extended substance," the predicate-term (extended substance) is, in fact, included in our concept of the subject (body) such that the proposition does not add anything to what the subject already contains. Locke, Berkeley, and Hume all devoted sections of their epistemological works to the relationship between words and things and to the fact that, very often, the only differences between things turn out to be a difference in the words we employ in describing them. The convention appearing in the eighteenth century was to label all such propositions in which the predicate is contained in the concept of the subject as "analytical." Kant retained this term and employed another to represent those propositions whose predicates are not logically implied by their subjects: that is, those propositions, that expand our factual knowledge. These he called "synthetic."<sup>60</sup> When we say that all bodies are heavy we are making an assertion about bodies different from what is contained in the mere concept we have of a body. The same is true of statements of the sort, "The French are a people of average height," "Protein is conducive to health," and so forth. Being French does not logically entail being average in height; being protein does not logically entail good health on the part of the recipient. In noting the general position of philosophers on the matter of analytic and synthetic propositions,



- be introduced (1990)

Kant formalized the distinction further by observing that the principle common to all analytic propositions is the law of contradiction.<sup>61</sup> We cannot say, "The man at the table is not the man at the table." In an affirmative analytic judgment, a contrary predicate will produce a contradiction. This is not the case with synthetic judgments, for no such contradiction results from, "The French are a people of above-average height." This distinction is what led all empirical philosophers, and Hume particularly, to subscribe to the view that analytical propositions are (a) logically necessary, meaning that if they are true, they must be true, (b) certain as opposed to probable, and (c) a priori as opposed to given in experience. Since  $a = a$  is necessarily true by the law of contradiction and since, on the empiricist account, nothing in experience is necessarily true,  $a = a$  is said to be known a priori. By the same token, the same philosophers insisted that synthetic propositions (a) can only be contingently and never necessarily true, (b) can only be assigned a certain probability of truth and never certainty, and (c) can only be advanced or assessed a posteriori. No rational, a priori deduction will establish beyond doubt that "The French people are of average height." In the light of these terms and distinctions, we can summarize Hume's position on morals, on epistemology, and on ethics by noting that he placed all these issues in the domain of synthetical judgments. Whatever we say about knowledge or values can be true only contingently, only some of the time, and only a posteriori. Kant's task, then, is to prove that some synthetic judgments are true a priori, and Kant's answer to Hume is, in essence, that there are a priori synthetic truths. Put another way, Kant's mission is to return necessity to morals and to epistemology and thereby rescue metaphysics from mere opinion.

Before turning to Kant's analysis, we must recognize that he stands in agreement with Hume on many counts. On one of the cardinal empiricist points, he, too, insists that all judgments of experience are synthetical,<sup>62</sup> that objects are given to us by means of the senses, and that thought itself, directly or indirectly, finally relates back to sensibility.<sup>63</sup> Kant, therefore, is not to be understood as one seeking to overturn empiricism but as one striving to determine its limits. It is in this regard that one might judge the empiricist movement as *culminating* in the *Critique of Pure Reason* rather than being swept aside by it.

Kant's critical analysis of Hume's claims predictably focuses on the Humean account of the concept of causation. The account is empirical, and Kant must determine the principles according to which experience yields the concept of cause. Alas, experience cannot yield the concept; experience assumes the concept. The proofs of this are contained in his famous *Analogies of Experience*: "The principle of the analogies is: *Experience is possible only through the representation of a necessary connection of perceptions*."<sup>64</sup> He presents three such analogies. The first is addressed to the concept of object-permanence. For an object or event to have any real existence, it must exist in time. Time, how-

1st  
World without order...  
...of C.E. ...  
...developed this notion...

ever, is not *given* to experience by the object or event. Only through our "inner intuition," in which permanence is cognized a priori, can appearances take place in time.<sup>65</sup> There can be no relation in time unless that relation is grounded in permanence. We discover the weight of smoke, for example, by weighing the wood, burning the wood, and weighing the ashes. Matter is not conserved in the senses; that is, the wood is now gone and no smoke can any longer be seen. Still, because of the a priori category of the understanding, the category of thought we call *permanence*, we know that the weight of the smoke is just this difference between the weight of the wood and the weight of the ashes.<sup>66</sup> We can only call an appearance a "substance" because we are able to presuppose the existence of substance throughout time. This brings Kant to the Second Analogy: "*Everything that happens, that is, begins to be, presupposes something upon which it follows according to a rule.*"<sup>67</sup> It is this cryptic principle that is the kernel of Kant's answer to Hume, and it must be studied carefully.

Hume had argued that our concept of cause was to be explained in terms of contiguity (spatial), constant conjunction, and succession. Briefly, A and B take place in the same location, always together, and in an invariant order, with A always preceding B. When these conditions are met, we say that "A is the cause of B." It is in the Second Analogy that Kant raises the question of the source of succession itself. We do not "see" time. We do not "perceive" intervals. We might *imagine* that a boat floating downstream *could* float upstream, but we do not apprehend the event in any but a fixed order: the boat now is here, next is there, next is there, and so forth. But what is the empirical basis of "next"? Quite simply, unless the understanding already possessed (a priori) the category *time*, there could be no succession or constant conjunction. Conjunctions occur in time, but time is not given by the object. For us to be affected by the constant conjunction of events A and B, we must be able to experience A and B as events. What makes them events is that they stand out against a background of enduring states of affairs. Against an enduring background of silence, for example, chimes are heard. Chimes can only be events if they are separable from the enduring background. And the chimes can only be caused by the striking of the hammer if our perceptions are unfailingly ordered in time. Without the a priori category of understanding—in contrast to sensation alone—we would have no basis for judging the hammer to cause the chimes any better than the basis for judging the chimes to cause the hammer. Yet, we never make the latter mistake. The ordering is not contingent, it cannot be a posteriori, and it is hardly merely probable.

Professor L. W. Beck has summarized the argument of the Second Analogy with an elegance and simplicity seldom displayed in the many treatments of "Kant's answer to Hume":



- K. Everything that happens, that is, begins to be, presupposes something upon which it follows by rule." (Kant's Second Analogy).
- P. Events can be distinguished from objective enduring states of affairs, even though our apprehension of each is serial (the accomplishment of Hume's task(a)).
- H. Among events, we find empirically some pairs of similar ones which tend to be repeated, and we then make the inductive judgment: events like the first members of the pairs are causes of events like the second (the accomplishment of Hume's task (b)).
- P implies K . . . H implies P, since if events cannot be distinguished, pairs of events cannot be found, and thus P is a necessary condition of H. Hence: H implies P and P implies K, therefore H implies K. That is Kant's answer to Hume.<sup>68</sup>

What Beck has shown is that Hume's account of causation requires, and logically requires in the sense of *necessarily requiring*, Kant's Second Analogy. Thus, Hume is not wrong, but he can be right only if the Second Analogy is granted and the Second Analogy confers on the understanding an a priori synthetic judgment.

The argument given in H (Hume's "succession" and "constant conjunction") can reduce the concept of causation to an inference from experience only by granting the percipient a basis for "first" and "next," and this basis is not, itself, grounded in experience but is assumed for experience to occur in the first place: that is, the Second Analogy. This is what Beck means by "H implies K." The Humean argument implies the Kantian argument.

Kant's analysis goes well beyond this question of the basis of causal inferences. The *Critique of Pure Reason* seeks to discover the foundations and principles of all knowledge, accepting at the outset that one foundation is, of course, the empirical. But, as the Humean account is insufficient to explain causation, Kant argues that the empirical account is insufficient to explain *anything* about human understanding except the conditions by which it becomes furnished with objects. What the understanding achieves is judgment. Judgment is based on logical functions. The latter are imposed upon the evidence of sense and are necessary prior to experience if experience is to have any meaning at all. These logical functions, which we possess intuitively, are the pure concepts of the understanding of which the following Table of Categories is exhaustive.<sup>69</sup>

#### I. Concept of Quantity:

Unity  
Plurality  
Totality

#### II. Concept of Quality:

Reality  
Negation  
Limitation

## III. Concept of Relation:

Inherence and Subsistence  
Cause and Effect  
Community

## IV. Concept of Modality:

Possibility-Impossibility  
Existence-Nonexistence  
Necessity-Contingency

These are the pure concepts of synthesis<sup>70</sup> which the understanding possesses a priori and without which coherent experience would be impossible. These categories contain the possibility of all experience in general.<sup>71</sup> We confront the world of sensibility with an understanding already possessed of such pure concepts as: either a thing exists, or it does not exist, or it exists in a limited way; either A is possible or it is impossible; either it happens to follow B or it must. We can frame universal propositions (All men are mortal) only if we possess intuitively the category of totality, and there is nothing in experience that can give this. That we arrive at the proposition by induction or generalization from a large number of cases is not to deny the category but merely to cite the conditions under which it is invoked. Obviously, to think of "all men" requires that we know of "men," and we can only come to know them through experience. But we can never know "all" of anything through experience. The very process of inference assumes the concept of quantity, and the very process of generalization within a class assumes the concept of relation.

The foregoing is Kant's epistemological argument against empiricism and, more important for Kant, it is the introduction to the moral argument against the empiricist's pleasure principle. There is no doubt but that if one subscribes to an empiricist epistemology one will have little patience with a moral science based on the "truths" of reason. Locke was willing to grant an axiomatic status to moral propositions, likening them to the propositions of geometry, but this truce with the rationalists was neither convincing nor long-lived. The Hume who cannot find necessity in the conjunction of natural events is hardly going to search for it in that conjunction of behavioral events we call moral conduct. Kant agrees that if epistemology were reducible to experience morality would be likewise. But having proved to his own satisfaction that epistemology cannot be reduced to the sensual domain and having established that this very world of sense is contained in the world of understanding, it must follow that the laws of experience are authored by those of reason.<sup>72</sup> Kant's metaphysics of morals thus is the crowning achievement of that rationalist morality to which Descartes, Spinoza, and Leibniz had devoted themselves. Moral precepts have the authority of reason not because they refer to something that does not occur in the real world but because our understanding of the real world is based on a rule without which understanding would be impossible. As the pure concepts of the understanding (i.e., the categories form the logical foundations upon which all our knowledge of the natural world is based) so too there is an a priori rational principle which makes moral judgments inescapable, universal in





form, and absolutely necessary to any explanation of the moral dimensions of life. It is not enough to argue that we judge "good" and "evil" on the basis of feeling unless one is able to explain why and how the given feeling attaches to the given act. The very attachment assumes a rule, and this rule is what Kant dubbed the "categorical imperative": act in such a way that the maxim of your action could serve as a universal law of nature.<sup>73</sup>

In its varied forms, the categorical imperative includes a reverence for law, the insistence that man is an end and never a means to some other end. The very concept of law assumes a rational animal who intends good effects.<sup>74</sup> The mere (empirical) listing of the observed consequences of actions will never disclose this intention, but the act itself could not have taken place had there not been the intention prior to it. To acknowledge this intention, the fact of intention, is to acknowledge simultaneously a freedom of the will. Its freedom is constrained in this sense: the very freedom requires that the will make law.<sup>75</sup> Reverence of the law is not acquired. The categorical imperative could not be acquired. The factual world of events could not be judged on a moral basis if there were not, a priori, pure moral concepts in the understanding. We are not "usually" or "contingently" ends in ourselves and not means toward some other end we have. We are necessarily ends in ourselves. We do not await the consequences of our actions in order to determine whether or not we *should* do unto others as we would have them do unto us. We understand that this is so or else we could never know guilt as long as we "got away" with what we did. There may be those who preach a situation-ethics, but they still draw the line at anarchy. And even those who might argue in favor of anarchy, if they are to argue at all, will begin with a principle and if that principle is ever to enjoy logical force, it will finally reduce to the categorical imperative—at which point, of course, it will contradict the claims of the anarchist.

Kant's influence in psychology has been far greater than is generally recognized. It is a commonplace in historical summaries to acknowledge his fame as a philosopher, to point to his nativistic emphasis and suggest its effect on some later psychologists. There are even those who have decided that Kant was really authoring a kind of anthropology and was anticipating subsequent instinct-theorists. He was decidedly not. It is, in fact, to his credit that he has influenced as many psychologists who have misunderstood him as he has those who have followed his reasoning. In subsequent chapters, we will have occasion to discuss theories of cognitive development, Gestalt psychology, genetic psychology, and moral development. We will review the ideas of Wundt, Freud, Köhler, and their disciples. In these subsequent chapters, it will become clear that, with the exceptions of behavioral and physiological psychology, there is no area of contemporary psychological concern that does not rely on the major elements of Kant's philosophy. Such topics as the innate, logical structure of thought and language, the a priori principles of perceptual organization, the

stages of cognitive and moral understanding, the concept of culture-neutral or culture-free methods of psychological assessment—these and many issues of lesser importance are hardly imaginable had Hume's sensationism prevailed to the exclusion of rationalism. Kant did not set out to save rationalism. Indeed, he had more admiration for Hume than for many of his detractors. He set out to establish the limits of knowledge and the conditions by which it takes place. In the process, he saved consciousness.

There were, however, several negative contributions contained in Kantian philosophy, at least as regards the appearance of experimental psychology. It was Kant's judgment that a bona fide science of the mind was something of a contradiction in terms. The mind, unlike external nature, does not stand still as we attempt to observe it. Indeed, the very attempt to observe its contents alters them. Moreover, what is most defining about the human mind are the a priori categories or pure understanding, and, as we have seen, these are not "given" by experience and do not have empirical content. Furthermore, they are necessary (rather than contingent or accidental) features of mind and, as such, are not reducible to biological or mechanical laws. Nothing in biology or machinery can be what it is necessarily, whereas the a priori categories are. Still, in his very pessimism, Kant did indirectly provide impetus to a psychology of perception, a study of consciousness:

Thus the whole of rational psychology, as a science surpassing all powers of human reason, proves abortive, and nothing is left for us but to study our soul under the guidance of experience, and to confine ourselves to those questions which do not go beyond the limits within which a content can be provided for them by possible inner experience.<sup>76</sup>

What he argued against in this passage was the notion that a deductive science of the human mind would extend our knowledge of the real world by unearthing those rational principles that regulate the real world. Such a view, in Kant's understanding, was taking to believing that we can increase the number of persons in a room by hanging mirrors on the wall! Instead he argues for a psychology that takes the contents of the mind as the only phenomena we can examine directly. The value of this psychology is of a negative sort—it permits us to criticize those productions of rationalism that collide with the facts of consciousness. On the whole, then, Kant's immense analysis imparted a conservative tone to experimental psychology. Writing in the long and broad shadow of the Kantian critical philosophy, Wundt would insist repeatedly that his psychology was nonmetaphysical, nondeductive, and concerned only with the facts and internal organization of consciousness. We shall examine further this part of the Kantian contribution in later chapters.



## The Rationalist Legacy

to fail something desirable

It does not detract from Descartes, Leibniz, Spinoza, and Kant to recall the extent to which seventeenth- and eighteenth-century rationalism echoed many of the major lessons of Plato, St. Augustine, and St. Thomas. The *Theaetetus* was a reply to Protagoras's empiricist contentions, and its argument is not too different from that offered by Leibniz in the *New Essays*. The agreement on substantial psychological issues between Spinoza and St. Augustine is too marked to require further comment. Kant was unique, but in several respects his uniqueness is to be traced to an antirationalist way of treating the problems. He was, for example, uncompromising in his opposition Berkeley's brand of idealism, he argued that mathematical judgments were synthetic, and he denied the existence of innate ideas, at least as these had been described in the rationalist tradition. His "transcendental aesthetic" located psychological principles above the plane of experience. The new experimental psychology of the nineteenth century would thus have difficulty finding a place for Kant and ultimately succeeded in doing this only by ignoring the rest of his philosophical system. Those who came to play the major part in the creation of psychology as an independent discipline strived to impart to the enterprise the same rigor and objectivity enjoyed by physics or mathematics. Where the model was physics, the rationalist tradition was an encumbrance, and where the model was mathematics the enterprise failed, or seemed to fail. Even those who adopted a Kantian perspective on the mind still found it necessary to employ the methods of the empiricist. The exception, of course, was Wundt, whose works will be reviewed later. Wundt sought the best of two worlds in attempting to build an empirical science on the foundation of (rational) introspection. We may judge his success by observing that there are not many Wundtians around any longer.

Had there been no other movement in philosophy and philosophical psychology, the twentieth-century psychologist would still be actively engaged in assessing Leibniz's answer to Locke, and Kant's answer to Hume. Argument and analysis would continue to be the methods of choice. However, even as the seventeenth- and eighteenth-century philosophers carried on their disputes, there was a mammoth enterprise unfolding, and at a rate that beggared the imagination of philosopher and layman alike: the enterprise of science, or what Bacon and, later, Newton called "experimental philosophy." In its stubbornly and innocently pragmatic way, it drew upon both empiricism and rationalism for support and inspiration, but wed itself to neither. In time, the empiricists would claim it, although no empirical philosopher of consequence ever made a contribution to it. Its real engines were skepticism and materialism: Descartes' method of doubt seeking assurance in technology. Its materialist foundations are the subject of the next chapter; of its skeptical component, enough has been said in this and the previous chapter.

What was the rationalist bequest? As we review the problems and methods of contemporary psychology, we find little direct evidence of a conscious commitment to the rationalist's vision. Research is addressed to behavioral engineering, brain physiology, social attitudes and influence, human learning and memory, and individual differences. Only the light writing addressed to a popular audience continues to concern itself with the "mind," seldom with the soul, and never with the monads. But when we move from research to theory, the picture changes. Many would agree that by 1970 the three most influential theoretical attempts or, shall we say, the three theoretical issues commanding the greatest attention involve (a) the stage-specific cognitive capacities of man during development from infancy, (b) the a priori faculties, which must be granted if human language is to be understood, and (c) the species-specific processes, which must be assumed if we are to account for the range of emotional, intuitive, and "moral" dispositions observed throughout the animal kingdom. While (c) received its direct impetus from Darwin, (a) and (b) stem unadorned from the rationalist tradition. It is only in the methods employed to investigate these issues that these contemporary concerns can be described as "empirical." They are rationalist issues at the core and they are the visible signs of the rationalist legacy.

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31. Ibid., Pt. III, Props. LIII, LV.
32. Ibid., Pt. VI, Prop. XI.
33. Ibid., Pt. II, Prop. XLVIII.
34. Ibid., Pt. III, Prop. II.
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42. Ibid.
43. Ibid.
44. Ibid.
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46. Ibid., #11.
47. Ibid., #19.
48. Ibid., #28.
49. Ibid., #20, 21.
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56. A remarkably brief and extraordinarily lucid summary of the Hume-Kant dispute

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62. Ibid.

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67. Ibid., A189.

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69. Kant, *Critique of Pure Reason*, B106.

70. Ibid.

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74. Kant, *Groundwork of the Metaphysic of Morals*, pp. 68–69.

75. Ibid., p. 98.

76. This passage is taken from A382 of the *Critique of Pure Reason*, which treats of the “paralogisms of pure reason.” Later (B421), Kant amplifies the usefulness of a rational psychology when it is developed as a discipline rather than as a doctrine: “It . . . keeps us, on the one hand, from throwing ourselves into the arms of a soulless materialism or, on the other hand, from losing ourselves in a spiritualism which must be quite unfounded so long as we remain in the present life.”



...they started having access to books, could do  
it more to learn about the body  
- Also got rid of God (could do so)

## 9 Materialism: The Enlightened Machine

### The Vexing Alternative - don't need soul

In the two preceding chapters, we have reviewed the philosophical psychologies of the English and Continental empiricists and rationalists, confining ourselves to the seventeenth and eighteenth centuries, when these become sufficiently definite to serve as alternatives. In Chapter 3, the very considerable similarities between Platonist and Aristotelian systems were noted. In subsequent chapters, equivalent similarities were found between philosophers identified with one or another school. In the Renaissance, and really not until the Renaissance, did we find signs of a clear break, the break between spiritualism and naturalism. But even in the Renaissance there were only signs, not dramatic ruptures. The great separation took place in the irresolvable conflicts between Hume's *Treatise* and Kant's *Critique*. Henceforth, empiricism and rationalism would hold out possibilities for radical departures and divisions. Unlike the differences that divided Greek atomists and Platonists, the modern antagonisms would surface as completed systems of thought, rich in implications and recommendations for social organization, law, morality, economics, and religion. By comparison, the disputes between orthodox Christians and Ockhamists in the thirteenth century or the Florentine Aristotelians and Platonists in the fifteenth would be nearly negligible.

As great as the break between rationalism and empiricism was, however, there remained ground common enough for a Kant even to attempt to answer a Hume. The philosophical disputes of the seventeenth and eighteenth centuries, to the extent that they were between rationalists and empiricists, were disputes about the same kinds of problems, and they were disputes that often began from identical starting points. Hume and Kant were both moved to explain the source of human knowledge, the nature of morality, and the character of society. Broadly, they were both engaged in the "science of mental life."